

Homework 2: The Atmosphere Solutions

Fall 2026

1 Spaceballs!

Question What is the total mass of the Earth's atmosphere? Use the MSIS profile posted on Canvas. If you crammed all of this atmosphere into a vacuum bag $10 \times 10 \times 100$ km in volume, at a temperature of 300 K, what would be the pressure in the bag? Assuming the molar mass for the air: 0.029 kg/mol.

Answer The MSIS profile provides mass density and height, the two pieces of information we need to solve for the mass. The following integral calculates the total mass of a sphere from the mass density $\rho(r)$:

$$m = \int \rho(r) dV$$

We can replace dV as follows:

$$V = \frac{4}{3}\pi r^3$$
$$dV = 4\pi r^2 dr$$

Using the example code in Matlab below, we find:

$$m = \int_{R_E}^{\infty} \rho(r) 4\pi r^2 dr \simeq \boxed{5.3 \times 10^{18} \text{ kg}}$$

Make sure you convert from the provided height (altitude) to the actual radius of the location by adding R_E to your altitude vector. Out of interest, note that this total mass is about *one millionth* (one part in 10^6) of the Earth's total mass.

```
1 z = 0:1:999; %km
2 MSIS = MSISatmosphere1000(z);
3 rho = MSIS.mass * 1e3; %g/cm3 to kg/m3
4 Re = 6378; %km
5 r = (z + Re) * 1e3; %m
6 integrand = rho.*4.*pi.*r.^2;
7 mass = trapz(r, integrand) %kg
```

For the pressure inside the vacuum, we need the ideal gas law, $PV = Nk_B T$ or $PV = nRT$. Let's use the $PV = nRT$ form and make a few substitutions:

$$PV = nRT = \frac{m}{M}RT \rightarrow P = \frac{m}{M} \frac{RT}{V}$$

where P is the pressure [Pa], V is the volume [$10 \times 10 \times 100 \text{ km} = 10^{13} \text{ m}^3$], n is the amount of substance of gas in moles [mol], R is the universal gas constant [$8.31 \text{ J mol}^{-1} \text{ K}^{-1}$], T is the temperature [300 K], m is the total mass of the gas [kg], and M is the molar mass [kg/mol]. While in reality, M would change with altitude, we can just use the value of M for dry air at the surface, 0.029 kg/mol, for the molecular mass inside the vacuum:

$$P = \frac{m}{M} \frac{RT}{V} = \frac{(5.3 \times 10^{18} \text{ kg})(8.31 \text{ J mol}^{-1} \text{ K}^{-1})(300 \text{ K})}{(0.029 \text{ kg/mol})(10^{13} \text{ m}^3)} \simeq \boxed{4.6 \times 10^{10} \text{ Pa}}$$

2 Atmosphere Scale Heights

Question Calculate the approximate scale heights for “air” (N_2/O_2 mixture at sea level), atomic oxygen, helium, and hydrogen, assuming a constant temperature of 300 K. Are these scale heights consistent with the profiles in Figure 1? Compare quantitatively, by calculating the scale height using the MSIS profiles again, at a few altitudes in the atmosphere appropriate to these three constituents. Explain qualitatively why the scale heights from these two methods differ.

Answer The formula for scale height is $H = \frac{kT}{Mg}$ where k is Boltzmann constant ($1.38 \times 10^{-23} \text{ J/K}$), T is the mean atmospheric temperature (300 K), M is the mean mass of the molecule (in kg), and g is the acceleration due to gravity at the surface (9.8 m/s^2). Only M is changing for each molecule:

$$\begin{aligned} M_{air} &= 28.96 \text{ u} & M_{He} &= 4.00 \text{ u} \\ M_O &= 15.99 \text{ u} & M_H &= 1.01 \text{ u} \end{aligned}$$

where u is the unified atomic mass unit equal to 1.66×10^{-27} . So just plug and chug!

$$\begin{aligned} H_{air} &= \frac{kT}{M_{air}g} \simeq \boxed{8.79 \text{ km}} & H_{He} &= \frac{kT}{M_{He}g} \simeq \boxed{63.6 \text{ km}} \\ H_O &= \frac{kT}{M_Og} \simeq \boxed{15.9 \text{ km}} & H_H &= \frac{kT}{M_Hg} \simeq \boxed{252 \text{ km}} \end{aligned}$$

We expect that when the molecular mass is smaller, the scale height will be larger, and thus the density profile will decrease more slowly with altitude.

We can estimate the scale height from the MSIS profile as the slope of height vs total number density in natural log-linear space. Since the number density in a given region can be written as $n(z) = n_0 \exp(-(z - z_0)/H)$, in natural log space we have $\ln(n/n_0) = -(z - z_0)/H$. We can take two points on the profile in a region where we want to estimate the scale height and use:

$$H = \frac{z_0 - z_1}{\ln(n_1) - \ln(n_0)}$$

Note that you have to use the natural log here, $\ln$, or else you'll have an extra factor of $\log_{10} e = 0.43$ in your scale height. With the MSIS profile from the previous problem, we can plot the total number density and select points from a few regions to estimate the slopes from Figure 2.

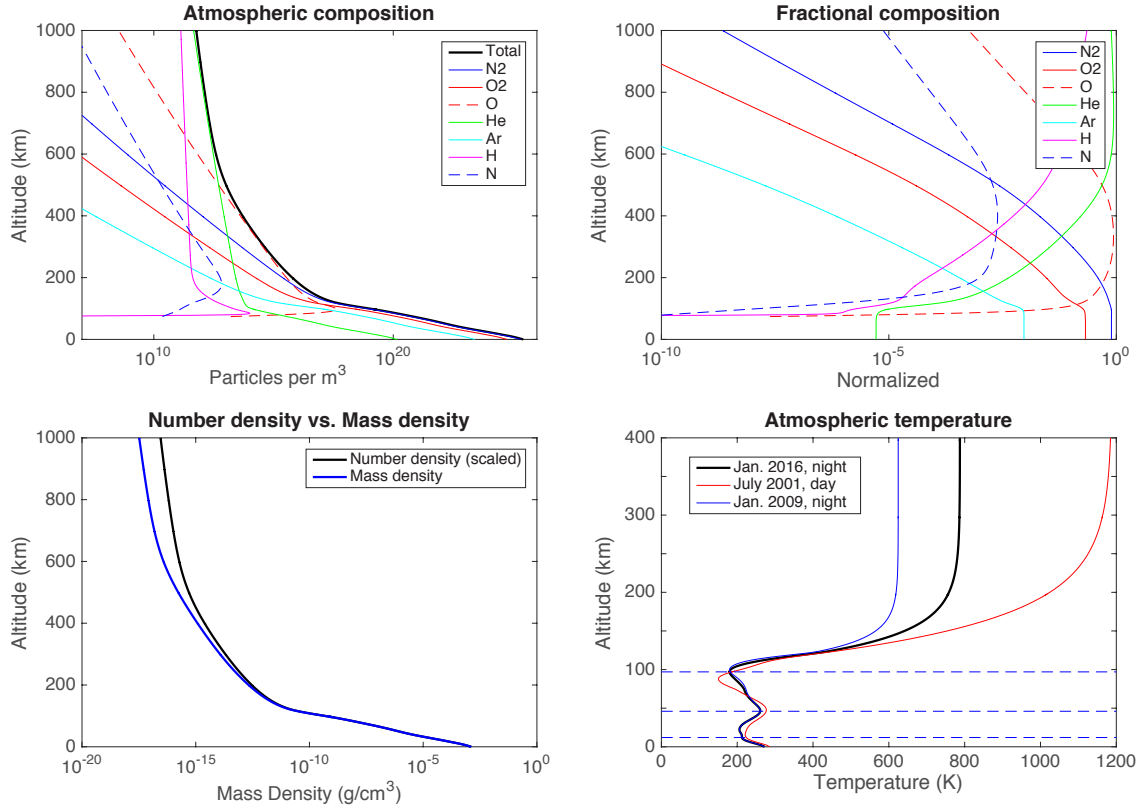


Figure 1: Atmospheric Profiles of (top left) composition, (top right) composition normalized to the total density, (bottom left) total mass density compared to number density, and (bottom right) temperature, for three different times.

Below about 100 km, our estimated scale height is close to the scale height we calculated for air:

$$H \sim \left| \frac{28 - 90}{54.48 - 45.49} \right| \simeq 6.90 \text{ km}$$

Between about 200 km and 500 km, the scale height is between what we calculated for oxygen and helium:

$$H \sim \left| \frac{203 - 378}{36.24 - 31.98} \right| \simeq 41.1 \text{ km}$$

Above about 600 km, the estimate is closest to what we calculated for hydrogen, but a bit lower:

$$H \sim \left| \frac{670 - 902}{28.55 - 27.49} \right| \simeq 217.7 \text{ km}$$

These estimates are only approximate; **the greatest error in our theoretical calculations was the assumption of a constant temperature of 300 K.** The differences between predicted and observed scale heights are largely due to this temperature assumption. In the middle region, for example, the temperature is more often closer to 1000 K or higher, which will increase the scale height by a factor of three or more. Gravity (g) is also decreasing with altitude, but that is a much smaller source of error.

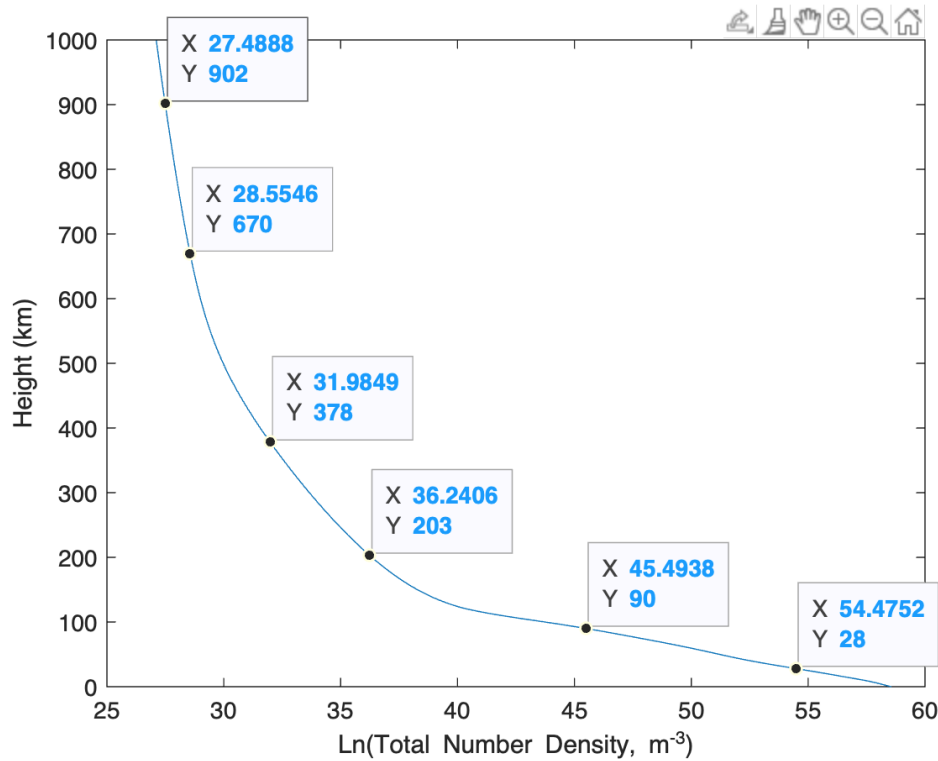


Figure 2: Estimating the scale heights in different altitude regions from a log-linear plot of the MSIS total number density.

You can get a more accurate estimate of the scale height at each altitude simply by taking the derivative of the log of the density. Since the scale height is related to the slope of the line in log space, this method will give you the local scale height at every altitude.

3 Atmospheric Variation

Question

- Use the [NRLMSISE-00 model online](#) to generate profiles of mass density versus altitude for solar minimum and solar maximum, up to 1000 km altitude. I suggest using 2001 for solar maximum, and 2009 for solar minimum; use January 1, 1200 UT, and 0 latitude and 0 longitude. Plot both profiles on a single axis and submit that figure with your assignment.
- What is the variation between solar maximum and solar minimum at 600 km altitude? What are the implications of this variation for spacecraft operations? (We're just looking for qualitative comments.)
- What is the dominant atmospheric constituent at 700 km during solar maximum? What about solar minimum?

Answer (i) The inputs we used for this solution key are shown in Figure 3.

After you generate your MSIS outputs for solar minimum and maximum, you need to convert the number densities (cm⁻³) to mass densities (g/cm³) before you plot. Simply multiply the number

NRLMSIS Atmosphere Model



Version
☐ NRLMSIS 2.0
☒ NRLMSIS-00

Time
Time Type
Coordinated Universal Time (UTC)
Date Time
01 / 01 / 2001 , 12 : 00 PM

Coordinate
Coordinate Type
Geographic
Latitude (-90° to 90°)
0
Longitude (0° to 360°)
0
Height (0 to 1000 km)
100

Profile
Profile Type
Height [0 to 1000 km]
Start
0
Stop
1000
Step Size
10

Optional Inputs
NOTE: -1 will use values from observational data sets for the selected time period
F10.7 (daily)
-1
F10.7 (3 month avg)
-1
AP (daily)
-1

Figure 3: MSIS inputs for solar maximum. Change the year to 2009 for solar minimum.

densities of each constituent by that constituent's molecular mass (e.g. $m_O = 15.999u \simeq 2.66 \times 10^{-23}$ g).

Now you can plot the MSIS mass density profiles on a single axis, and you should have your x-axis log-scaled. Your plot should look like Figure 4, where the dashed lines correspond to solar minimum and the solid lines correspond to solar maximum.

(ii) Here are the differences from solar min to solar max ($\rho_{max} - \rho_{min}$), at 600 km altitude, for each constituent (you could also report these as fractional changes, rather than absolute changes):

$$\begin{aligned}
 \text{O} &: 3.57 \times 10^{-16} \text{ g/cm}^3 \\
 \text{N}_2 &: 7.30 \times 10^{-18} \text{ g/cm}^3 \\
 \text{O}_2 &: 6.03 \times 10^{-20} \text{ g/cm}^3 \\
 \text{He} &: 5.43 \times 10^{-18} \text{ g/cm}^3 \\
 \text{Ar} &: 1.21 \times 10^{-22} \text{ g/cm}^3 \\
 \text{H} &: -2.34 \times 10^{-19} \text{ g/cm}^3 \\
 \text{N} &: 1.28 \times 10^{-17} \text{ g/cm}^3
 \end{aligned}$$

From calculating these differences or looking at your plot, you can see that most of the constituents have much higher mass density during solar max than solar min. However, H shows lower mass density at solar max than solar min at 600 km.

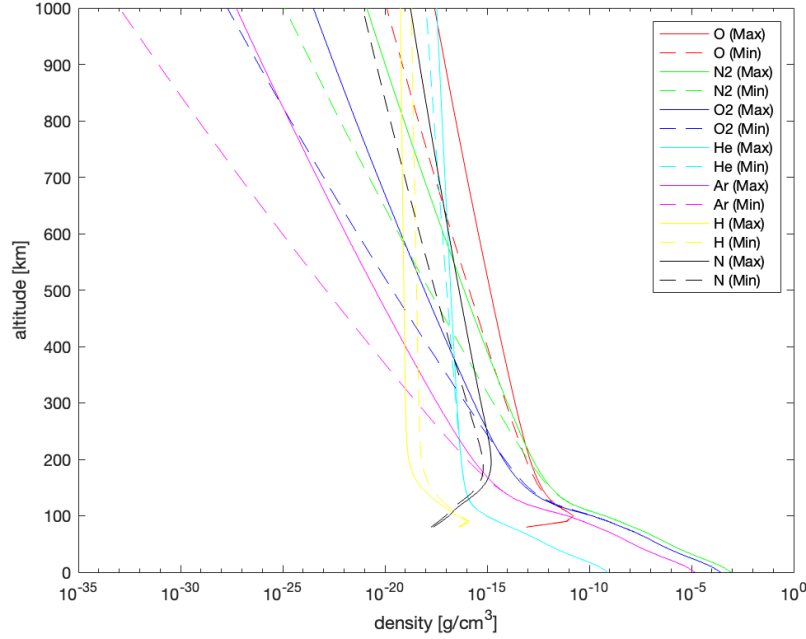


Figure 4: Mass densities for solar minimum and solar maximum conditions.

(iii) Looking at our plot, the dominant constituent at 700 km during solar maximum is oxygen (solid red line) and during solar minimum is helium (dashed cyan line).

4 Oxygen Impact

Question If the temperature in the thermosphere is 1000 K, i) what is the characteristic velocity of an oxygen atom (O)? ii) How much energy (in eV) is imparted to a spacecraft in LEO for each collision with an O atom? iii) What fraction of the atomic oxygen population has sufficient energy to cause sputtering on Aluminum surfaces? (See Figure 15 above. Consider just the 1D velocities parallel to the spacecraft ram direction.)

Answer (i) Find the mean velocity $\langle v \rangle$ or the RMS velocity v_{rms} of an O atom with $T = 1000$ K, $m = 15.999$ u, and $u = 1.66054 \times 10^{-27}$ kg:

$$\langle v \rangle = \sqrt{\frac{8k_B T}{\pi m}} \simeq \boxed{1.150 \text{ km/s}}$$

$$v_{\text{rms}} = \sqrt{\langle v^2 \rangle} = \sqrt{\frac{3k_B T}{m}} = 1.085 \langle v \rangle \simeq \boxed{1.250 \text{ km/s}}$$

(ii) Since the thermal velocity is much less than the spacecraft velocity (~ 7.8 km/s in LEO), the kinetic energy is mostly from the spacecraft velocity:

$$E = \frac{1}{2} m v_{sc}^2 = \frac{1}{2} (2.66 \times 10^{-26} \text{ kg}) (7.8 \times 10^3 \text{ m/s})^2 \simeq 8.09 \times 10^{-19} \text{ J} \simeq \boxed{5.04 \text{ eV}}$$

You can obtain a more accurate energy range from including the mean (rms) speed of the oxygen atom:

$$E_{max} = \frac{1}{2}m(v_{sc} + v_O)^2 \simeq 6.64 \text{ eV (6.79 eV)}$$

$$E_{min} = \frac{1}{2}m(v_{sc} - v_O)^2 \simeq 3.67 \text{ eV (3.55 eV)}$$

(iii) To start, we will find the threshold velocity of atomic oxygen, with the target material of aluminum. From Figure 15, an energy of 23 eV is needed to cause sputter, so first convert that energy to Joules using the conversion factor of $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$:

$$E_t = 23 \text{ eV} \cdot 1.6 \times 10^{-19} \text{ J/eV} = 3.68 \times 10^{-18} \text{ J}$$

Using this threshold energy and the mass of atomic oxygen, we can obtain the threshold velocity:

$$v_t = \sqrt{\frac{2 \cdot E_t}{m_0}} = \sqrt{\frac{2 \cdot 3.68 \times 10^{-18}}{16 \cdot 1.67 \times 10^{-27}}} \simeq 16.6 \text{ km/s}$$

This threshold velocity tells us how fast the atom is moving *relative to the spacecraft* to cause sputtering. But the spacecraft is not stationary; it is moving at the orbital velocity at the given altitude of 600 km:

$$v_{orbit} = \sqrt{\frac{GM}{R_E + h}} = \sqrt{\frac{3.986 \times 10^{14}}{6370 \times 10^3 + 600 \times 10^3}} \simeq 7.56 \text{ km/s}$$

Since we are making the simplifying assumption of a 1D problem, where atoms traveling perfectly parallel (or anti-parallel) to ram direction, the real threshold will be $v_t - v_{orbit}$, or approximately 9 km/s. To find the fraction of particles with at least this velocity, we need to integrate the Maxwell-Boltzmann distribution in 1D from this relative threshold to infinity. The Maxwell-Boltzmann distribution is given by:

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-\frac{mv^2}{2k_B T}}$$

where for this problem $T = 1000 \text{ K}$. Note that only particles with velocities above the approximately 9 km/s threshold will cause sputtering. To find the fraction of particles with at least this speed, we have to integrate the Maxwell-Boltzmann distribution. The trapz function in MATLAB is recommended for simplicity. An example code is shown below:

```

1 % Problem 4.iii:
2 %
3 % If the temperature in the thermosphere is 1000 K,
4 % iii) What fraction of the atomic oxygen population has sufficient energy
5 % to cause sputtering on Aluminum surfaces? (See Figure 15 above.
6 % Consider just the 1D velocities parallel to the spacecraft ram direction.)
7
8 clear; close all;
9
10 % constants

```

```

11 kb = 1.38e-23;
12 mO = 16 * 1.67e-27;      % Mass of atomic oxygen with m = 16u
13 GM = 3.986e14;           % GM constant; m^3/s^2
14 Re = 6370e3;             % Earth radius
15
16 % parameters
17 T = 1000;                % Temperature at 600 km
18 h = 600e3;               % altitude
19
20 % let's start by finding the threshold velocity.
21 Et = 23 * 1.6e-19;       % 23 eV converted to Joules
22 vt = sqrt(2*Et/mO);      % Gives a threshold of about 16.6 km/s
23
24 % What is the orbit velocity at 600 km?
25 v_orbit = sqrt(GM/(Re+h));
26
27 % so the real threshold is actually vt - v_orbit, about 9 km/s
28 v_thresh = vt - v_orbit;
29
30 % Now, let's plot a Maxwell-Boltzmann distribution, in 1D, covering at
31 % least up to those velocities.
32 v = linspace(0.1e3,20e3,2000);
33
34 f = 4*pi * (mO/(2*pi*kb*T)) ^ 1.5 * (v.^2) .* exp(-(mO*v.^2)/(2*kb*T));
35
36 figure(1);
37 plot(v/1e3,f);
38 set(gca,'FontSize',16);
39 xlabel('Velocity (km/s)')
40 ylabel('f(v) (Normalized)')
41 title('Maxwell-Boltzmann distribution for O at 1000 K')
42
43 hold on;
44 plot([v_thresh v_thresh]/1e3,get(gca,'ylim'),'r:');
45
46 % Now I want to integrate: the entire distribution, and then just the part
47 % above v_thresh. Use trapz for simplicity.
48 A = trapz(v,f);
49
50 inds = find(v >= v_thresh);
51 B = trapz(v(inds),f(inds));
52
53 frac = B/A;               % This gives a number around 10^-34!
54
55 % BUT consider that half the particles are going the other direction in
56 % this 1D example; so the true fraction is half of this answer
57
58 realfrac = frac/2

```

The above script gives a wildly small answer, on the order of 10^{-34} ! Note that since we are assuming a 1D system (essentially just a line), the atoms can go in two directions: opposite to ram or parallel to ram. Particles have an equal probability of traveling in either direction relative to the spacecraft, so only half of the particles above the threshold velocity are traveling towards the spacecraft, and therefore with the higher relative speed of 16.6 km/s.. Therefore, we have to divide the integrated answer in half. Doing this results in the even smaller answer of 2.23×10^{-34} !

We can conclude from this problem that there are most likely very few atomic oxygen particles that are moving fast enough to cause sputtering on an aluminum spacecraft, even at 1000 K. Indeed, Oxygen Attack is a far more important process than sputtering for oxygen.

5 Orbit Decay

Question i) Derive equation:

$$\frac{dP}{dt} = -3\pi a \rho \frac{C_D A}{m} \quad (1)$$

that relates the change in orbit period to the mass density of the atmosphere.

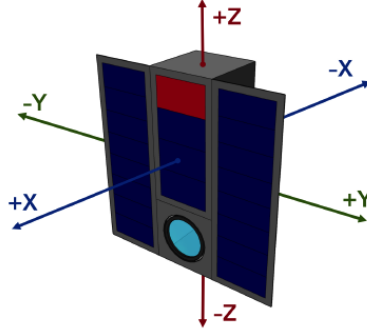


Figure 5: CubeSat for problem 5. The CubeSat is $10 \times 10 \times 30$ cm, and the solar panels are 10×30 cm.

ii) The satellite in Figure 5 with $C_D = 2.2$ is in orbit at 400 km altitude. If the satellite is flying with its ram direction in the $+x$ direction shown, how long until it deorbits (reaching at 120km)? Again, use the MSIS profile posted on Canvas for the mass density of the atmosphere. iii) Repeat for a starting altitude of 500 km. iv) Repeat for a starting altitude of 400 km again, but now with the spacecraft flying in the $+z$ direction. For all of these cases, make plots of the satellite altitude vs. time!

Answer (i) For a circular orbit, $r = a$ in all equations. First, we combine the orbital period and velocity equations:

$$\begin{aligned} P^2 &= \frac{4\pi^2}{GM_e} r^3 & v &= \sqrt{\frac{GM}{r}} \\ &= \frac{4\pi^2}{GM_e} \left(\frac{GM}{v^2} \right)^3 & r &= \frac{GM}{v^2} \\ &= \frac{4\pi^2 G^2 M^2}{v^6} \\ P &= \frac{2\pi GM}{v^3} \end{aligned}$$

Next, we take the time derivative of P and combine with the drag force:

$$\begin{aligned}
\frac{dP}{dt} &= 2\pi GM \frac{d}{dt}(v^{-3}) \\
&= 2\pi GM(-3v^{-4}) \frac{dv}{dt} \\
&= 2\pi GM(-3v^{-4}) \frac{\rho v^2 C_D A}{2m} \\
&= -3\pi \rho \frac{GM}{v^2} \frac{C_D A}{m} \\
&= -3\pi \rho r \frac{C_D A}{m}
\end{aligned}
\qquad
\begin{aligned}
F &= m \frac{dv}{dt} = \frac{1}{2} \rho v^2 C_D A \\
\frac{dv}{dt} &= \frac{\rho v^2 C_D A}{2m} \\
\frac{GM}{v^2} &= r \text{ (from above)}
\end{aligned}$$

We made it to the orbit decay equation!

(ii-iv) As described in Section 4.1 of the Atmosphere course notes, we need to iterate through these equations to determine how the orbit changes with time:

$$\begin{aligned}
P^2 GM_e &= 4\pi^2 a^3 \\
\frac{dP}{dt} &= -3\pi a \rho \frac{C_D A}{m}
\end{aligned}$$

where P is the orbital period [s], G is the gravitational constant [$6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$], M_e is the mass of the Earth [$5.97 \times 10^{24} \text{ kg}$], a is the orbital radius [m], ρ is the density the spacecraft is flying through [kg/m^3], C_D is the drag coefficient, m is the spacecraft mass [kg], and A is the spacecraft cross-sectional area [m^2].

For each step in the iteration, we also need to recalculate the density, ρ , at the new altitude. Surface area, A , changes for the two different directions. When flying in the $+x$ direction, $A = A_{\text{SC-side}} + 2A_{\text{solar-panel}} = (10 \text{ cm} \times 30 \text{ cm}) + 2(10 \text{ cm} \times 30 \text{ cm}) = 900 \text{ cm}^2 = 0.09 \text{ m}^2$. When flying in the $+z$ direction, $A = A_{\text{SC-top}} = 10 \text{ cm} \times 10 \text{ cm} = 100 \text{ cm}^2 = 0.01 \text{ m}^2$.

Here is an example in MATLAB of what the iteration may look like:

```

1 % orbit decay
2
3 clear all;
4
5 % constants
6 G = 6.67e-11;
7 Re = 6378000;
8 Me = 5.98e24;
9
10 % variables
11 A = (0.3*0.3); % spacecraft cross section area, m2
12 h0 = 500e3; % initial altitude, m
13 M = 4; % satellite mass, kg
14 Cd = 2.2;
15
16 alt = h0;
17 deorbit = 120e3;
18 timestep = 60*60*3*1; % increment in seconds; this is three hours
19 tstepdays = 60*60*24;
20
21 steps = 5000;
22
23 vel = zeros(1,steps);

```

```

24 alt = h0*ones(1,steps);
25 time = zeros(1,steps);
26
27 period(1) = sqrt(4*pi^2 * (Re+alt(1))^3 / G / Me);
28 period(2:steps) = 0;
29
30 for i = 1:steps,
31
32     if (alt(i) < deorbit),
33         break;
34     end
35
36     nd = MSISatmosphere1000(alt(i)/1e3);
37     DN = nd.mass * 1e3;
38
39     %SH = (900 + 2.5 * (ften - 70) + 1.5 * Ap) / (27 - .012 * (alt/1e3 - 200));
40     %DN = 6e-10 * exp(-(alt/1e3 - 175) / SH);
41
42     % change in period
43     dP = 3 * pi * A / M * (Re+alt(i)) * DN * Cd * timestep;
44     period(i+1) = period(i) - dP;
45     alt(i+1) = (period(i+1)^2 * G*Me / (4*pi^2))^(1/3) - Re;
46
47     time(i+1) = i*timestep;
48
49     vel(i+1) = sqrt(G*Me/(Re+alt(i+1)));
50
51     fprintf('Day %.2f: Altitude %.2f km\n',i*timestep/tstepdays,alt(i+1)/1e3);
52
53 end
54
55 timey = time/60/60/24/365.25;
56
57 figure;
58 ax1 = subplot(131);
59 plot(ax1,timey*365,period/60);
60 xlabel(ax1,'Time (years)');
61 ylabel(ax1,'Orbit period (minutes)');
62
63 ax2 = subplot(132);
64 plot(ax2,timey,alt/1000);
65 xlabel(ax2,'Time (years)');
66 ylabel(ax2,'Orbit altitude (km)');
67
68 ax3 = subplot(133);
69 plot(ax3,timey,vel/1000);
70 xlabel(ax3,'Time (years)');
71 ylabel(ax3,'Orbit velocity (km/s)');

```

For the initial altitude of 400 km and $+x$ direction: deorbit in 167 days.

For the initial altitude of 500 km and $+x$ direction: deorbit in 1360 days.

For the initial altitude of 400 km and $+z$ direction: deorbit in 1502 days.

All plots are shown in Figure 6.

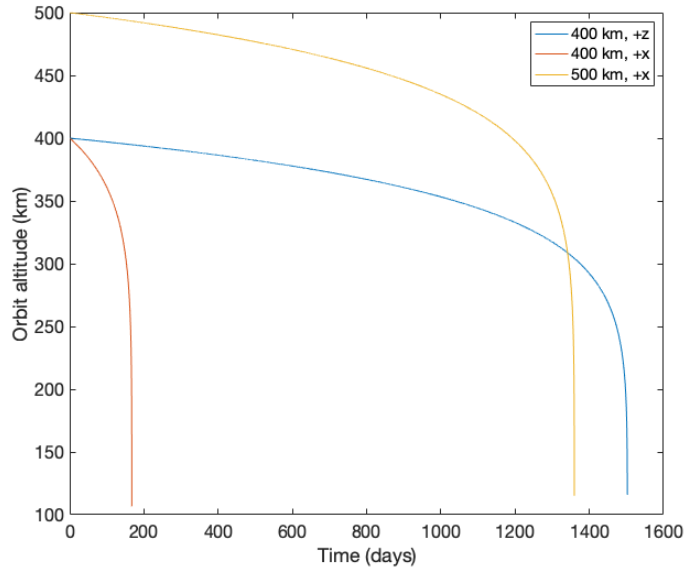


Figure 6: Altitude vs. time profiles for the three deorbiting cases in Problem 5.

6 Atmosphere Uncertainty and Neutral Winds

Question Let's add some uncertainties in our calculations for the problem above. Consider the first case from above, where our spacecraft is at 400 km altitude, flying in the $+x$ spacecraft direction.

- i) Now assume there is an atmosphere density uncertainty of 10%. Repeat the calculation from the previous problem. Plot the altitude vs. time, showing both cases (this one and Problem 5) on the same plot. What is the difference in deorbit time?
- ii) Given that the orbit around the Earth is roughly 90 minutes (give or take) at any LEO altitude, what are the implications for where on Earth this spacecraft will re-enter the atmosphere?
- iii) BONUS: Repeat the calculation, but assume a constant neutral wind of 100 m/s directed towards the spacecraft ram face (i.e, a headwind). [*This is hard! You have to re-derive the equation that you are iterating; why?? If you don't get to this part, don't worry about it.*]

Answer (i) This is a simple modification; just multiply your mass density by 0.9 or 1.1 every time it is used. (The tricky part is writing elegant matlab code that iterates over the various parameters...)

Results are plotted in Figure 7. This 10% uncertainty in the mass density results in orbit decay lifetimes that vary by about 10%. Specifically, these are the values we get:

Case	90% Density	Nominal Density	110% Density
400 km, $+x$	186 days	167 days	152 days
500 km, $+x$	1510 days	1360 days	1236 days
400 km, $+z$	1669 days	1502 days	1366 days

(ii) You'll notice that the decay time with this 10% uncertainty in the atmosphere leads to a decay time difference much greater than one orbit. Hence, even a much smaller uncertainty than this

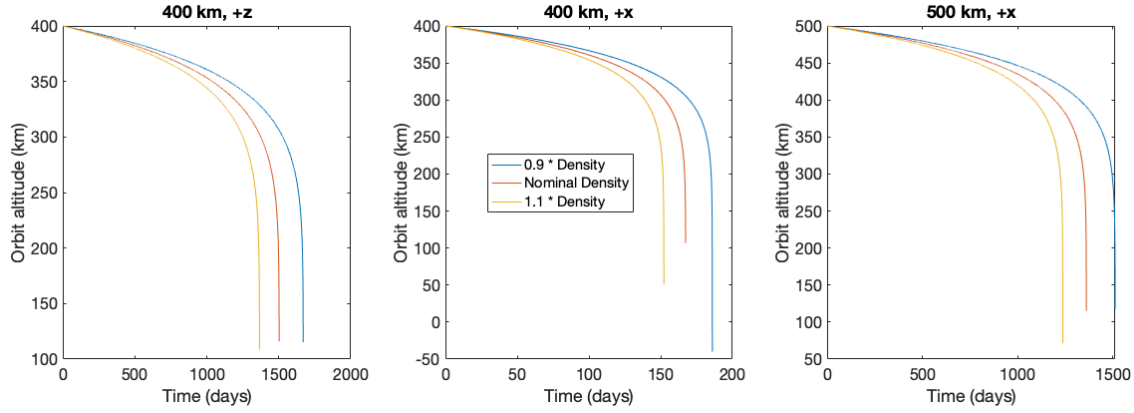


Figure 7: Altitude vs. time profiles for the three deorbiting cases in Problem 5, now with $\pm 10\%$ uncertainty in mass density.

will lead to a position error larger than the Earth's circumference, meaning it will be impossible to predict where the spacecraft will re-enter.

In reality, we track objects and update their orbital parameters frequently. For example, we tracked CIRBE every day, mostly for the S-band downloading. S-Band has a narrow angle range, and the ground station has to point to the s/c accurately to obtain efficient reception. Once the altitude drops below 300 km, it is very difficult to predict the position of the s/c. It will be much harder to predict the reentry time and location unless we can also track the s/c constantly, e.g., every hr or shorter.

(iii) (Bonus part) The addition of a wind velocity changes the set of equations above and requires a new derivation. The velocity of the spacecraft in Kepler's third law, v , does not change; but the relative velocity between the spacecraft and the atmosphere, $v_A = v + w$, which is relevant to the drag force, does change due to the wind velocity w . (This is the answer to the *why??* posed in the problem statement.)

We can skip some steps in the solution to Problem 5 above and start in the middle. The drag force is now:

$$F = m \frac{dv}{dt} = \frac{1}{2} \rho (v + w)^2 C_D A \quad \rightarrow \quad \frac{dv}{dt} = \frac{\rho (v + w)^2 C_D A}{2m}$$

and the equation for dP/dt becomes:

$$\begin{aligned} \frac{dP}{dt} &= 2\pi GM(-3v^{-4}) \frac{dv}{dt} \\ &= 2\pi GM(-3v^{-4}) \frac{\rho (v + w)^2 C_D A}{2m} \\ &= -3\pi \rho \frac{GM}{v^2} \frac{C_D A}{m} \frac{(v + w)^2}{v^2} \\ &= -3\pi \rho r \frac{C_D A}{m} \frac{(v + w)^2}{v^2} \quad \text{(using } \frac{GM}{v^2} = r \text{ as before)} \end{aligned}$$

Now there is this extra factor of $(v + w)^2/v^2$ compared to earlier. You can add this factor directly in your code where you calculate dP at each time step (being careful to use the current orbital velocity v at each step).

Note that this is a relatively small contribution; with $v \simeq 7.7$ km/s and $w = 0.1$ km/s, this factor equals 1.026. Nonetheless, it will affect the orbit propagation. When I plug this into my Matlab code for the de-orbit time for each of our three cases, this is what I get:

Case	$w = 0$ m/s	$w = 100$ m/s
400 km, +x	167 days	163 days
500 km, +x	1360 days	1325 days
400 km, +z	1502 days	1464 days

Of course, in reality, the wind will not be constant, nor in a constant direction; the true effect on a spacecraft orbit has to account for the changing wind speed and direction over its lifetime (which, of course, we cannot currently measure).

7 Air-breathing Thruster

Question Imagine a spacecraft at 400 km that lets in the neutral atmosphere through an inlet, ionizes it, and accelerates it out the back. The spacecraft has $C_D = 2.2$ and $A = 100$ cm². i) What thrust is required to balance the force of drag? ii) What is the exhaust velocity required for the atomic oxygen to achieve this thrust? (hint: $\Delta m/\Delta t$, the exhaust mass rate, must be the same as the mass collection rate)

Answer (i) We balance the drag force with a thrust of equal magnitude in the opposite direction. To find the drag force, $F_D = \frac{1}{2}\rho v^2 C_D A$, we need to first find v :

$$\begin{aligned}
 v &= \sqrt{\frac{GM_E}{r}} = \sqrt{\frac{GM_E}{h + R_E}} \\
 &= \sqrt{\frac{(6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2})(5.97 \times 10^{24} \text{ kg})}{400000 \text{ m} + (6.371 \times 10^6 \text{ m})}} \simeq 7,670 \text{ m/s}
 \end{aligned}$$

We can find ρ at 400 km from the MSIS profile to be 1.194×10^{-12} kg m⁻³ (answers that use just the density of atomic oxygen are acceptable since this thruster is designed for ionizing oxygen). Now we just plug the values in!

$$\begin{aligned}
 F_D &= \frac{1}{2}\rho v^2 C_D A \\
 &= \frac{1}{2}(1.194 \times 10^{-12} \text{ kg m}^{-3})(7670 \text{ m s}^{-1})^2(2.2)(0.01 \text{ m}^2) \\
 &\simeq \boxed{7.73 \times 10^{-7} \text{ N}}
 \end{aligned}$$

(ii) The thrust force is related to the exhaust velocity through the following equation:

$$F_{thrust} = v_e \frac{\Delta m}{\Delta t} \rightarrow v_e = \frac{F_{thrust}}{\Delta m/\Delta t}$$

where v_e is the effective exhaust velocity measured in m/s and $\frac{\Delta m}{\Delta t}$ is the exhaust mass rate in kg/s. To find $\frac{\Delta m}{\Delta t}$, we need to find the mass collection rate, from our hint! We can find this as follows:

$$\frac{\Delta m}{\Delta t} = \rho A v = (1.194 \times 10^{-12} \text{ kg m}^{-3})(0.01 \text{ m}^2)(7670 \text{ m s}^{-1}) \simeq 9.16 \times 10^{-11} \text{ kg s}^{-1}$$

This assumed that the mass collection area is equal to the given cross sectional area and that the mass is collected at the same speed as the spacecraft's orbital velocity. Adding this all together:

$$v_e = \frac{F_{thrust}}{\Delta m / \Delta t} \simeq \boxed{8.44 \text{ km/s}}$$

This required exhaust velocity is actually lower than a typical ion thruster; more commonly the exhaust velocity is in the low tens of km/s. This lends some credence to the feasibility of this air-breathing thruster concept for drag make-up.

8 Missions

Question How do we measure the number density, and/or the composition, of the thermosphere? Research one of the following methods. Describe how the method works and describe an instrument on a past or current mission (or on the ground) that makes these measurements.

- Space-based Ion and Neutral Mass Spectrometer (INMS)
- Ground-based incoherent scatter radar (ISR)
- Space-based accelerometer measurements (e.g., CHAMP)
- Another method that you discover in your literature search

Answer Varies by selected method. Slack us if you have questions!